

WASP-35b

R-1119

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-35_241218 · created 2026-07-30T21:38:28+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460662.7022 +/- 0.0036 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.105 +/- 0.011
Transit depth (Rp/Rs)^2	1.11 +/- 0.23 [%]
Orbital Inclination (inc)	89.78 +/- 1.14
Airmass coefficient 1 (a1)	1.2541 +/- 0.009
Airmass coefficient 2 (a2)	-0.177 +/- 0.014
Scatter in the residuals of the lightcurve fit is	0.71 %
Best Comparison Star	None
Optimal Aperture	4.01
Optimal Annulus	11.37
Transit Duration (day)	0.0851 +/- 0.004

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.105 ± 0.011 vs 0.1241 (deviation 1.55σ)
Duration fit vs published	0.0851 d vs 0.1317 d (deviation 10.4σ)
Mid-time O–C	-5.0 min
Depth SNR	8.5
Residual scatter	0.71 %

Light curve

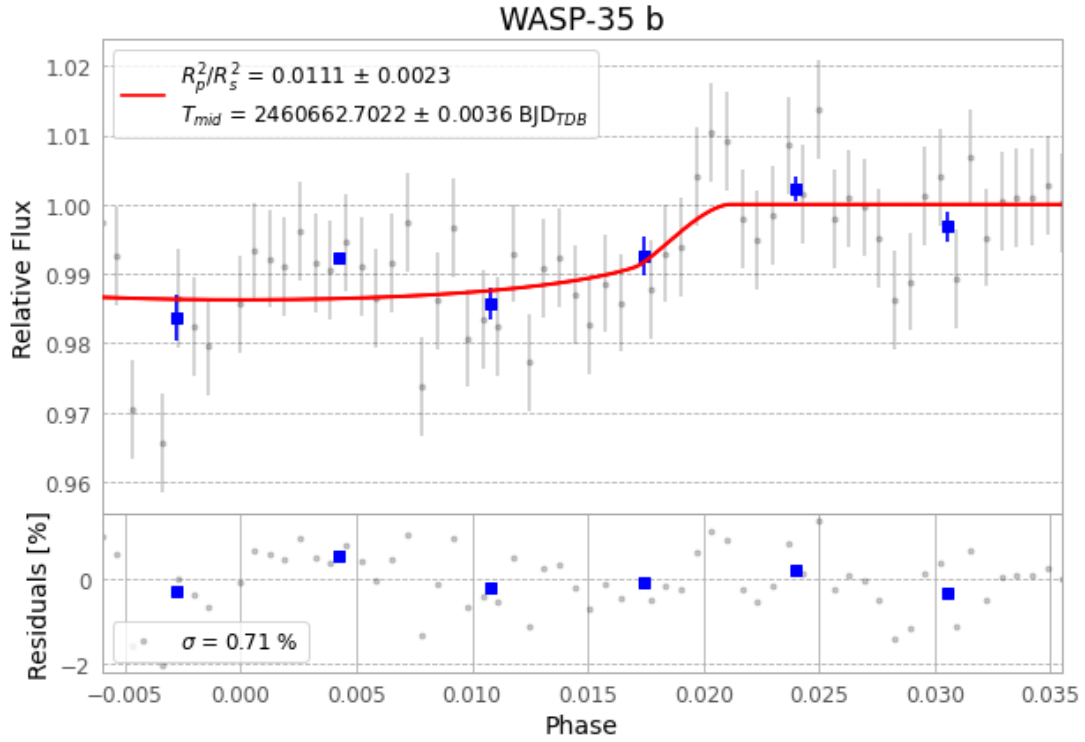


Figure 1 — FinalLightCurve_WASP-35 b_2024-12-17.png (SHA-256 592b86f4eea17c96...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [374, 204], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 7fe05ad7611e7550...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

64 FITS frames (42 MB), WASP-35241218041516.FITS → WASP-35241218072415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-35-241218.log (b5949f97c62c...), FinalLightCurve_WASP-35 b_2024-12-17.png (592b86f4eea1...), FinalLightCurve_WASP-35 b_2024-12-17.csv (cb03716a2b36...)

Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 21:38Z.